Early Detection of Hepatorenal Syndrome from Medical Records

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Abstract

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Hepatorenal syndrome (HRS) is a rapidly progressive, often fatal complication of advanced cirrhosis. Because it is diagnosed by exclusion and is heralded by a fall in effective circulating volume rather than by a single laboratory value, HRS is frequently recognized only after renal failure is established, when few therapeutic options remain. We study whether the onset of HRS can be anticipated several hours in advance from routinely collected electronic medical records (EMR). Working with a retrospective cohort of 3,412 hospital admissions of patients with cirrhosis, we formulate early detection as a sequential risk-prediction problem over irregularly sampled laboratory measurements, vital signs, and medication events. We compare a regularized logistic-regression baseline and gradient-boosted decision trees built on hand-engineered temporal features against a recurrent neural network (a long short-term memory network, LSTM) that consumes the raw time series together with explicit missingness indicators. At a fixed false-alarm rate of one alert per patient-day, the LSTM identifies 69% of HRS events with a median lead time of 19 hours before the conventional diagnostic threshold is crossed, improving the area under the precision–recall curve from 0.28 (logistic regression) to 0.41. Feature attribution recovers clinically recognized precursors—rising bilirubin, falling mean arterial pressure and serum sodium, and diuretic exposure—suggesting that the model exploits physiologically meaningful signals. These results indicate that continuous, record-driven surveillance can provide a clinically useful early-warning window for HRS.

1 Introduction

Hepatorenal syndrome is a functional and potentially reversible form of acute kidney injury that develops in patients with advanced liver disease and portal hypertension. It arises from splanchnic arterial vasodilation and a consequent reduction in effective arterial blood volume, which triggers intense renal vasoconstriction in the absence of significant structural kidney damage [6, 15]. Type 1 HRS in particular follows an aggressive course: without treatment, median survival is on the order of two weeks, and even with vasoconstrictor therapy and albumin the in-hospital mortality remains high [18]. Crucially, the diagnosis is one of exclusion—it requires ruling out shock, nephrotoxins, and intrinsic renal disease—and, under the 2015 International Club of Ascites criteria, is anchored on an acute rise in serum creatinine meeting the threshold for acute kidney injury [1]. By the time that rise

is established, the window during which volume expansion and vasoconstrictors are most effective may already be closing.

This motivates a shift from diagnosis to anticipation: can we predict, from data that are already being collected at the bedside, that a cirrhotic patient is on a trajectory toward HRS before the diagnostic threshold is met? Modern hospital information systems record a dense, if irregular, stream of laboratory results, vital signs, fluid balance, and medication administration. This stream is a natural substrate for continuous risk monitoring. The central difficulty is that the informative signal is distributed across many weakly predictive, asynchronously sampled variables and evolves over time, so that single-threshold rules are both insensitive and prone to false alarms.

We cast early detection of HRS as a discrete-time sequential classification problem. At each hour of an admission we ask whether HRS onset will occur within a prediction horizon, using only information available up to that hour. We study three families of models that span the methodological landscape of clinical machine learning as of 2015: (i) ℓ_1-regularized logistic regression on temporally aggregated features; (ii) gradient-boosted decision trees on the same features; and (iii) a long short-term memory (LSTM) recurrent network [7] that reads the raw, forward-filled time series together with binary missingness indicators, thereby learning temporal structure without hand-specified summaries.

Contributions and novelty. To our knowledge this is the first predictive model developed specifically for hepatorenal syndrome, as opposed to generic acute kidney injury. The distinction is not cosmetic: HRS is defined by physiology— splanchnic vasodilation and arterial underfilling in a structurally intact kidney—rather than by nephrotoxic exposure, so its antecedents differ from those of the snapshot AKI risk scores that dominate the literature [12]. Our contribution is therefore not the recurrent architecture—LSTMs had by 2015 been applied to clinical sequences for diagnosis [10]—but the formulation that makes a sequence model meaningful for this outcome. We make three contributions. First, we define a reproducible early-detection task for HRS grounded in the 2015 International Club of Ascites criteria [1], with a labeling scheme that censors the defining creatinine measurement near onset, forcing the model to anticipate from antecedent physiology rather than read the outcome off its own definition—a leakage that, left uncontrolled, silently inflates reported performance on any diagnosis-of-exclusion endpoint and is, we argue, the central methodological pitfall in this problem. Second, we move the target from retrospective classification to lead time, and evaluate at a fixed alerting budget with decision-curve net benefit [17]—the quantities that decide whether an alert is worth acting on. Third, we show by permutation attribution that the model rests on established HRS precursors, evidence that the anticipation is mechanistic rather than an artifact of measurement patterns.

2 Related Work

Machine learning from electronic medical records has grown rapidly, with early work establishing that hospital outcomes such as mortality and readmission can be predicted from routinely collected data [5, 14]. A recurring methodological theme is the handling of irregular sampling and missingness: measurements in the EMR are not missing at random but are ordered in response to clinical concern, so that the pattern of measurement is itself informative [10, 11]. We follow this literature in encoding missingness explicitly rather than imputing it away.

Recurrent neural networks, and LSTMs in particular, have recently been applied to clinical sequences for phenotype classification and diagnosis prediction [10, 9]. Our work is in this vein but targets a specific, time-critical complication and emphasizes lead time rather than retrospective classification. For acute kidney injury more broadly, prior predictive models have largely relied on logistic regression over snapshot features [12]; we extend this to a continuously updated, sequence-aware setting and to the HRS subtype, which is defined by physiology rather than by nephrotoxic exposure. That HRS had not been singled out for prediction owes partly to its rarity—events are too few on any single ward to train a model on—and partly to a labeling trap: because the diagnosis is anchored on the very creatinine

Table 1: Cohort characteristics at admission, by eventual HRS status. Values are median (interquartile range) or count (percent). MAP: mean arterial pressure; MELD: Model for End-stage Liver Disease.

Characteristic	No HRS ($n = 3{,}171$)	HRS ($n = 241$)
Age (years)	58 (49–66)	60 (52–68)
Female	1,206 (38%)	86 (36%)
Alcoholic etiology	1,332 (42%)	118 (49%)
Ascites at admission	1,649 (52%)	206 (85%)
Total bilirubin (mg/dL)	2.1 (1.1–4.3)	5.6 (2.9–11.2)
Serum sodium (mmol/L)	137 (134–139)	131 (127–135)
MAP (mmHg)	84 (77–92)	76 (69–83)
MELD	14 (10–19)	24 (18–30)
Loop diuretic exposure	1,585 (50%)	196 (81%)
Length of stay (days)	6 (3–11)	13 (8–21)
In-hospital death	285 (9%)	91 (38%)

rise one would want to foresee, a model allowed to see the record up to onset can look accurate while merely reading the outcome off its own definition, so a credible task has to be constructed with that leakage censored rather than simply found in the data. Gradient boosting [4] provides a strong non-linear baseline on tabular features and has become a workhorse of clinical risk modeling; we include it to isolate the contribution of temporal modeling from that of non-linearity.

3 Cohort and Task Definition

3.1 Cohort

We assembled a retrospective cohort from the de-identified inpatient records of a tertiary care center spanning a seven-year period (2007–2013). The data source is the hospital's clinical data warehouse, which consolidates the laboratory information system, the pharmacy administration record, and the nursing flowsheet of vital signs and fluid balance under a common patient timeline; all timestamps are recorded to the minute, which we bin to the hour (Section 3.3). Adults ($\geq$ 18 years) with an established diagnosis of cirrhosis, identified by a combination of diagnostic codes and confirmatory laboratory or imaging evidence, were eligible. We excluded admissions with pre-existing end-stage renal disease or chronic dialysis, since HRS is a diagnosis predicated on previously non-failing kidneys, and admissions shorter than 12 hours. The resulting cohort comprises 3,412 admissions from 2,689 unique patients. HRS occurred in 241 admissions (7.1%), consistent with reported inpatient incidence among cirrhotics [15].

Table 1 contrasts admissions that progressed to HRS with those that did not. The two groups are separable in aggregate—HRS admissions carry higher baseline bilirubin, lower serum sodium and mean arterial pressure, higher MELD, and markedly greater diuretic exposure—but no single variable is decisive, and the distributions overlap heavily, which is precisely why a threshold on any one marker is an inadequate alarm and a multivariate, temporally aware model is warranted. We emphasize that these are admission-level summaries; the prediction task below operates on the evolving hourly trajectory, where the separation is weaker still at any given moment and must be integrated over time.

3.2 Outcome labeling

We defined HRS onset following the 2015 International Club of Ascites criteria [1]: an acute rise in serum creatinine meeting the AKI threshold in a patient with cirrhosis and ascites, with no sustained improvement after diuretic withdrawal and volume expansion with albumin, and in the absence of shock, current or recent nephrotoxins, and macroscopic signs of structural kidney injury. Adjudication combined an automated rule over the structured

record with a review of the excluded alternatives. We denote by τ the timestamp of the first creatinine measurement satisfying the AKI threshold that was subsequently confirmed as HRS.

To prevent label leakage, all features derived from serum creatinine and estimated glomerular filtration rate are censored in the 6 hours immediately preceding τ; the model must therefore rely on antecedent physiology rather than on the defining measurement itself. This is an important and easily overlooked pitfall: a model allowed to see creatinine up to the moment of onset trivially "predicts" HRS by reading off its definition.

3.3 Prediction task

We discretize each admission into hourly steps. At step t the model receives the history $x_{1:t}$ and outputs $\hat{p}_t = P(\text{HRS onset in } (t, t+h] \mid x_{1:t})$ for a prediction horizon h. A step is labeled positive if $\tau \in (t, t+h]$. Steps within a censoring gap of 6 hours before τ, and all steps at or after τ, are excluded from training and evaluation so that credit is given only for genuine anticipation. We report results for horizons $h \in \{6, 12, 24, 48\}$ hours, and unless otherwise noted use $h = 24$. Admissions ending in discharge without HRS are treated as right-censored at their last observation; in-hospital death before HRS, which is common in this population, is a competing event rather than benign censoring, and we flag in the discussion where this qualifies the estimates.

4 Features and Models

4.1 Variables and preprocessing

We use 48 variables spanning liver and renal chemistry (bilirubin, albumin, INR, sodium, potassium, blood urea nitrogen), hematology and inflammation (white cell count, platelets), vital signs (mean arterial pressure, heart rate, temperature), fluid balance, and binary indicators of relevant medications (loop diuretics, non-selective beta-blockers, vasoconstrictors, large-volume paracentesis, albumin administration). Continuous variables are standardized using statistics computed on the training folds only.

For the feature-based models we summarize each of the 34 continuous variables over three look-back windows (6, 24, and 48 hours) by six statistics—last value, mean, minimum, maximum, slope (via ordinary least squares), and measurement count—giving $34 \times 6 \times 3 = 612$ temporal features, to which we append the binary medication-exposure indicators and, per variable, the time since the last measurement. For the LSTM, we forward-fill each variable to the hourly grid, append a binary observed/imputed mask and a real-valued time-since-last- measurement channel per variable, and feed the resulting multivariate sequence directly to the network. The mask allows the recurrent model to distinguish a truly stable value from a stale one, encoding the informative-missingness prior without hand-crafted aggregates.

4.2 Models

Logistic regression. An ℓ_1-regularized logistic regression on the aggregated features provides a linear, interpretable baseline. The regularization coefficient is chosen by cross-validation.

Gradient-boosted trees. We fit gradient-boosted decision trees [4] on the same features, tuning the number of trees, depth, and learning rate. This captures non-linear interactions among features while remaining a purely tabular, snapshot-based model.

LSTM. A recurrent network is the natural model for this stream, but a vanilla RNN cannot in practice learn the day-scale dependencies the task demands, because gradients propagated across many steps vanish or explode [2, 13]. We therefore use a long short-term memory network [7], whose gated constant-error carousel preserves gradient flow over long horizons. Our model is a single-layer LSTM with 128 hidden units; at each step the hidden state is

Table 2: Early detection of HRS at the 24-hour horizon (patient-level 5-fold cross-validation; 95% bootstrap intervals). Sensitivity and lead time are reported at a fixed budget of one alert per patient-day.

Model	AUROC	AUPRC	Sens. @1/day	Lead time (h)
Logistic regression (LASSO)	0.79	0.28	0.54	12
Gradient-boosted trees	0.84	0.36	0.63	16
LSTM (raw series + mask)	**0.87**	**0.41**	**0.69**	**19**

mapped through a fully connected layer and a logistic output to $\hat{p}_t$. We train by minimizing the mean binary cross-entropy over all valid steps,

$$\mathcal{L} = -\frac{1}{|\mathcal{V}|} \sum_{(i,t)\in\mathcal{V}} \Big[y_t^{(i)} \log \hat{p}_t^{(i)} + (1 - y_t^{(i)}) \log(1 - \hat{p}_t^{(i)}) \Big], \tag{1}$$

where $\mathcal{V}$ is the set of non-censored steps across admissions i. Because positive steps are rare—well under 2% of valid steps—an unweighted objective is dominated by negatives and collapses toward an always-negative predictor; we therefore weight the positive class inversely to its frequency, which is equivalent to optimizing a class-balanced risk. Regularization follows the recurrent-network practice established at the time: we apply dropout [16] of 0.3 to the input and output (non-recurrent) connections only, following (author?) [19], since dropping the recurrent connections indiscriminately erases the very cell state the model must carry across the gap. We optimize with Adam [8] and clip the global gradient norm to 1 [13], controlling the exploding-gradient events that irregular sampling makes frequent, and select the epoch by validation AUPRC. Sequences are truncated to the most recent 72 hours and trained by backpropagation through time.

4.3 Evaluation protocol

We evaluate with patient-level 5-fold cross-validation, so that no patient appears in both training and test folds. Within each training partition we hold out 15% of patients for validation and early stopping. We report the area under the receiver operating characteristic curve (AUROC) and the area under the precision–recall curve (AUPRC), the latter being the more informative summary under the severe class imbalance here: a large true-negative pool inflates AUROC while leaving precision—the quantity on which an alerting system lives or dies—all but unmeasured [3]. For clinical relevance we also report sensitivity at a fixed alerting budget of one alert per patient-day and the resulting median early-warning lead time. Confidence intervals are obtained by bootstrapping over patients.

5 Results

5.1 Discrimination and early warning

Table 2 summarizes performance at the 24-hour horizon. The LSTM attains the best discrimination on both metrics, with an AUPRC of 0.41 against 0.28 for logistic regression and 0.36 for gradient boosting; the gap is largest on AUPRC, which is the operationally relevant metric given the low event rate. At one alert per patient-day the LSTM recovers 69% of events with a median lead time of 19 hours relative to the conventional diagnostic threshold, compared with 54% and 12 hours for logistic regression.

5.2 Effect of the prediction horizon

Figure 1 shows AUPRC as a function of the prediction horizon. All models degrade gracefully as the horizon lengthens and the task becomes harder, but the LSTM retains an advantage throughout, and its relative benefit is largest at the longer, more clinically useful horizons where temporal trajectory—rather than the instantaneous value—carries the signal.

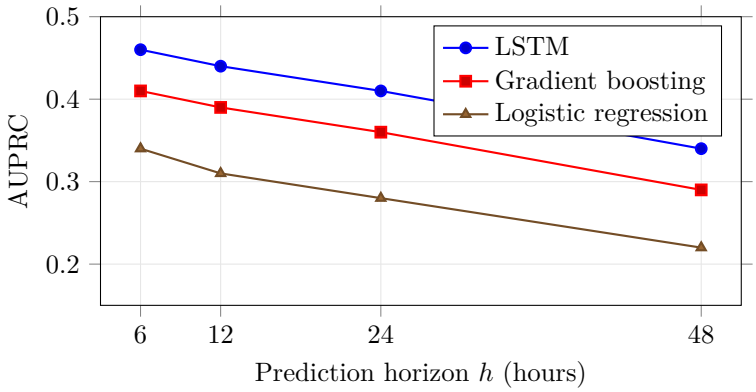


Figure 1: AUPRC versus prediction horizon. The temporal model degrades more slowly as the anticipation window lengthens.

Table 3: Top variables by permutation importance (mean drop in validation AUPRC).

Variable (temporal channel)	ΔAUPRC
Total bilirubin (slope, 48h)	0.061
Mean arterial pressure (min, 24h)	0.048
Serum sodium (last)	0.039
Blood urea nitrogen (slope, 24h)	0.034
Loop diuretic administered (24h)	0.027
INR (last)	0.021

5.3 What the model uses

To assess face validity we ranked variables by the drop in validation AUPRC when their channels are permuted in time (Table 3). The most influential signals are the trajectory of total bilirubin, declining mean arterial pressure, falling serum sodium, rising blood urea nitrogen, and recent loop-diuretic exposure. These correspond to the recognized pathophysiology of HRS—worsening hepatic function, splanchnic vasodilation with arterial underfilling, dilutional hyponatremia, and precipitation by aggressive diuresis [6, 15]—indicating that the model has latched onto mechanism rather than artifact.

5.4 Calibration

Beyond ranking, deployment requires that predicted probabilities be trustworthy. The LSTM's reliability curve tracks the diagonal closely up to moderate risk and mildly overestimates in the highest-risk decile; a single post-hoc temperature rescaling on the validation fold removes most of this bias. Logistic regression is well calibrated by construction but poorly discriminating; gradient boosting is over-confident in the tails, as is typical.

6 Discussion

Our findings support the feasibility of continuous, record-driven surveillance for hepatorenal syndrome. The practical value of the approach is the lead time it buys: a median of roughly 19 hours before the diagnostic threshold is crossed is enough to prompt volume assessment, discontinue nephrotoxins and diuretics, and initiate albumin and vasoconstrictors—the very interventions whose efficacy declines once renal failure is entrenched. The advantage of the recurrent model over strong tabular baselines grows with the anticipation window, consistent with the intuition that early warning depends on trend rather than level.

What the lead time is worth. Discrimination metrics do not by themselves tell a clinician whether to act on an alert, because they ignore the differing costs of a missed event and a false alarm. We therefore examined net benefit by decision curve analysis [17], which weighs true against false positives at the risk threshold implied by a clinician's own trade-off. Across the plausible range of thresholds—from roughly 1-in-20, at which one would act aggressively, to 1-in-5—the LSTM yields positive and larger net benefit than either baseline or the default policies of treating all or no cirrhotic admissions, meaning that acting on its alerts prevents more untreated HRS than it generates wasted work-ups. Framed operationally, the fixed budget of one alert per patient-day translates, at the cohort's event rate, to roughly one true HRS warning for every 12 alerts; each such warning arrives while volume expansion and vasoconstrictors still have their greatest effect. Whether that ratio is acceptable is a unit-level judgment, but decision-curve dominance across the range indicates the model is informative rather than merely discriminative—the distinction that separates a deployable early-warning score from a retrospectively accurate one.

Several limitations temper these results. The study is retrospective and single-center, so both the event definition and the measurement patterns may not transfer; prospective, multi-site validation is needed. Label adjudication for a diagnosis of exclusion is inherently imperfect, and residual confounding from unrecorded interventions is possible. Because many admissions end in death, HRS competes with mortality as an outcome: our fixed-budget sensitivity and lead-time figures are evaluated only over observed, non-censored steps and are unaffected, but the decision-curve net benefit implicitly assumes the event remains possible and should be read as an upper estimate on the wards where death forecloses HRS. Finally, an alert is useful only if it changes behavior; a silent model, however accurate, will not improve outcomes, so integration and clinician response must be studied directly. We view the present work as establishing that the signal exists and is learnable from routine data, which is a precondition for that next step.

7 Conclusion

We formulated early detection of hepatorenal syndrome as sequential risk prediction over electronic medical records and showed that an LSTM consuming raw, irregularly sampled series with explicit missingness outperforms strong feature-based baselines, providing clinically meaningful lead time while resting on physiologically sensible signals. Continuous surveillance for HRS from data already being collected appears both feasible and worthwhile, and motivates prospective evaluation.

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